In God we trust

نام و نام خانوادگی : مهدیس میرزایی

شماره دانشجویی : ۸۱۰۱۰۲۵۵۰

شماره پروژه : پروژه سوم درس

قسمت اول:از اولین Thread تا ساخت و رفع Race Condition

بخش a :

در این قسمت صرفا مرحله به مرحله پیش رفتیم و خروجی به صورت زیر شد :

```
❯ gcc task_1a.c -o task_1a -pthread
❯ ./task_1a
Enter the number of threads : 10
Thread 1 of 10 starting. Sleeping for 1 seconds...
Thread 2 of 10 starting. Sleeping for 1 seconds...
Thread 3 of 10 starting. Sleeping for 1 seconds...
Thread 4 of 10 starting. Sleeping for 1 seconds...
Thread 5 of 10 starting. Sleeping for 2 seconds...
Thread 6 of 10 starting. Sleeping for 3 seconds...
Thread 7 of 10 starting. Sleeping for 2 seconds...
Thread 8 of 10 starting. Sleeping for 3 seconds...
Thread 9 of 10 starting. Sleeping for 2 seconds...
Thread 10 of 10 starting. Sleeping for 3 seconds...
I'm thread 1 of 10
I'm thread 2 of 10
I'm thread 3 of 10
I'm thread 4 of 10
I'm thread 5 of 10
I'm thread 7 of 10
I'm thread 9 of 10
I'm thread 6 of 10
I'm thread 8 of 10
I'm thread 10 of 10
```

و ترتیب چاپ پیام‌ها غیرقابل پیش‌بینی است زیرا **زمان‌بندی CPU توسط سیستم عامل انجام می‌شود**. این زمان‌بندی تابع عواملی چون Preemption، تصادفی بودن Sleep و Interleaving اجرای دستورات است. در نتیجه، سیستم عامل می‌تواند هر thread را در هر لحظه متوقف کرده و thread دیگری را اجرا کند، و این امر ترتیب دقیق خروجی را نامشخص می‌سازد

بخش b :

```
❯ gcc task_1b.c -o task_1b -pthread
❯ ./task_1b
Single-Thread Sum for N=10000:
10000 0
Single-Thread Sum for N=1000000:
1000000 1
Single-Thread Sum for N=5000000:
5000000 6
Single-Thread Sum for N=10000000:
10000000 10
Single-Thread Sum for N=50000000:
50000000 55

T=2 T=4 T=10 T=100 T=1000
Multi-Thread Sum for N  =10000
0 0 0 1 15
Multi-Thread Sum for N  =1000000
1 0 1 1 12
Multi-Thread Sum for N  =5000000
3 2 1 1 15
Multi-Thread Sum for N  =10000000
5 3 3 2 15
Multi-Thread Sum for N  =50000000
28 14 11 11 17
```

همانطور که مشاهده می‌شود افزایش تعداد thread ها تا حدی خوب است, به عبارتی از تعدادی به بعد هزینه ساخت آنها از کاربردشان بیشتر است و باعث کند شدن نسبت به حالت‌هایی که تعداد thread ها کمتر بوده است می‌شود.

بخش c :

```
❯ gcc task_1c1.c -o task_1c1 -pthread
❯ ./task_1c1
Race Condition Scenario
Expected Counter Value: 400000
Actual Counter Value: 124498
Difference: 275502
Race Condition Scenario
Expected Counter Value: 4000000
Actual Counter Value: 1025989
Difference: 2974011
❯ gcc task_1c2.c -o task_1c2 -pthread
❯ ./task_1c2

Mutex Safe Scenario
Expected Counter Value: 400000
Actual Counter Value: 400000
Difference: 0

Mutex Safe Scenario
Expected Counter Value: 20000000
Actual Counter Value: 20000000
Difference: 0
```

همانطور که مشاهده می‌شود با اضافه کردن mutex این مشکلات برطرف شد و مقداری که انتظار داشتیم با پاسخ به دست آمده یکی شد.

قسمت دوم :شبیه‌ساز زمانبندی CPU: FCFS، SJF، Round Robin

خروجی سناریو اول :

```
First Scenario:

FCFS

Gantt Chart:
| P1 | P2 | P3 | P4 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 8     | 0     | 8      | 0       | 8          |
| P2      | 0       | 4     | 8     | 12     | 8       | 12         |
| P3      | 0       | 1     | 12    | 13     | 12      | 13         |
| P4      | 0       | 3     | 13    | 16     | 13      | 16         |

Average Waiting Time: 8.25
Average Turnaround Time: 12.25

SJF Non-Preem

Gantt Chart:
| P3 | P4 | P2 | P1 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 8     | 8     | 16     | 8       | 16         |
| P2      | 0       | 4     | 4     | 8      | 4       | 8          |
| P3      | 0       | 1     | 0     | 1      | 0       | 1          |
| P4      | 0       | 3     | 1     | 4      | 1       | 4          |

Average Waiting Time: 3.25
Average Turnaround Time: 7.25

Round Robin, (q=1)

Gantt Chart:
| P1 | P2 | P3 | P4 | P1 | P2 | P4 | P1 | P2 | P4 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 8     | 0     | -1     | 0       | -1         |
| P2      | 0       | 4     | 1     | -1     | 1       | -1         |
| P3      | 0       | 1     | 2     | 3      | 2       | 3          |
| P4      | 0       | 3     | 3     | 10     | 3       | 10         |

Average Waiting Time: 1.50
Average Turnaround Time: 2.75

Round Robin, (q=4)

Gantt Chart:
| P1 | P2 | P3 | P4 | P1 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 8     | 0     | 16     | 0       | 16         |
| P2      | 0       | 4     | 4     | 8      | 4       | 8          |
| P3      | 0       | 1     | 8     | 9      | 8       | 9          |
| P4      | 0       | 3     | 9     | 12     | 9       | 12         |

Average Waiting Time: 5.25
Average Turnaround Time: 11.25
```

خروجی سناریو دوم :

```
Second Scenario:

FCFS

Gantt Chart:
| P1 | P2 | P3 | P4 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 7     | 0     | 7      | 0       | 7          |
| P2      | 2       | 4     | 7     | 11     | 5       | 9          |
| P3      | 4       | 1     | 11    | 12     | 7       | 8          |
| P4      | 5       | 4     | 12    | 16     | 7       | 11         |

Average Waiting Time: 4.75
Average Turnaround Time: 8.75

SJF Non-Preem

Gantt Chart:
| P1 | P3 | P2 | P4 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 7     | 0     | 7      | 0       | 7          |
| P2      | 2       | 4     | 8     | 12     | 6       | 10         |
| P3      | 4       | 1     | 7     | 8      | 3       | 4          |
| P4      | 5       | 4     | 12    | 16     | 7       | 11         |

Average Waiting Time: 4.00
Average Turnaround Time: 8.00

Round Robin, (q=1)

Gantt Chart:
| P1 | P1 | P2 | P1 | P2 | P3 | P1 | P4 | P2 | P1 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 7     | 0     | -1     | 0       | -1         |
| P2      | 2       | 4     | 2     | -1     | 0       | -3         |
| P3      | 4       | 1     | 5     | 6      | 1       | 2          |
| P4      | 5       | 4     | 7     | -1     | 2       | -6         |

Average Waiting Time: 0.75
Average Turnaround Time: -2.00

Round Robin, (q=4)

Gantt Chart:
| P1 | P2 | P3 | P1 | P4 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 7     | 0     | 12     | 0       | 12         |
| P2      | 2       | 4     | 4     | 8      | 2       | 6          |
| P3      | 4       | 1     | 8     | 9      | 4       | 5          |
| P4      | 5       | 4     | 12    | 16     | 7       | 11         |

Average Waiting Time: 3.25
Average Turnaround Time: 8.50
```

خروجی سناریو سوم :

```
Third Scenario:

FCFS

Gantt Chart:
| P1 | P2 | P3 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 20    | 0     | 20     | 0       | 20         |
| P2      | 0       | 3     | 20    | 23     | 20      | 23         |
| P3      | 0       | 3     | 23    | 26     | 23      | 26         |

Average Waiting Time: 14.33
Average Turnaround Time: 23.00

SJF Non-Preem

Gantt Chart:
| P2 | P3 | P1 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 20    | 6     | 26     | 6       | 26         |
| P2      | 0       | 3     | 0     | 3      | 0       | 3          |
| P3      | 0       | 3     | 3     | 6      | 3       | 6          |

Average Waiting Time: 3.00
Average Turnaround Time: 11.67

Round Robin, (q=1)

Gantt Chart:
| P1 | P2 | P3 | P1 | P2 | P3 | P1 | P2 | P3 | P1 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 20    | 0     | -1     | 0       | -1         |
| P2      | 0       | 3     | 1     | 8      | 1       | 8          |
| P3      | 0       | 3     | 2     | 9      | 2       | 9          |

Average Waiting Time: 1.00
Average Turnaround Time: 5.33

Round Robin, (q=4)

Gantt Chart:
| P1 | P2 | P3 | P1 | P1 | P1 | P1 |

| Process | Arrival | Burst | Start | Finish | Waiting | Turnaround |
|---------|---------|-------|-------|--------|---------|------------|
| P1      | 0       | 20    | 0     | 26     | 0       | 26         |
| P2      | 0       | 3     | 4     | 7      | 4       | 7          |
| P3      | 0       | 3     | 7     | 10     | 7       | 10         |

Average Waiting Time: 3.67
Average Turnaround Time: 14.33
```